

FPGA-Based EO-PLL With Repetitive Control for Highly Linear Laser Frequency Tuning in FMCW LIDAR Applications

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Abstract—We developed a digital electro-optical phase-locked-loop (EO-PLL) utilizing a repetitive control, allowing the use of very nonlinear tunable lasers for frequency-modulated continuous-wave (FMCW) light detection and ranging (LIDAR) applications. This approach allows reducing the phase error of continuously repeated frequency chirps to virtually zero by using information of previous chirps and additionally guarantees sub-second settling times. With this method, much higher nonlinearities in the laser frequency modulation slope γ are tolerable than with common real-time analog PLLs. Furthermore, we applied a Goertzel algorithm generalized to non-integer multiples of the fundamental frequency to achieve a mean measurement precision of 5.41 μm and 16.35 μm at a distance from the interferometer baseline of 3 m and 10.3 m, respectively.

Index Terms—Electro-optical phase-locked-loop (EO-PLL), frequency-modulated continuous-wave (FMCW) LIDAR, Goertzel algorithm, range finding, repetitive control, velocimetry.

I. INTRODUCTION

FREQUENCY-modulated continuous-wave (FMCW) LIDAR appears more and more to be a capable or even superior alternative to conventional radar or light detection and ranging (LIDAR) systems for automotive or industry applications. Due to the superior resolutions achievable with frequency-modulated continuous-wave (FMCW) LIDAR, it is also suitable for high precision 3D-scanning applications. Main advantages of the FMCW approach are robustness against interference and much lower necessary laser peak powers, what also improves eye safety. A further advantage is the additional Doppler velocity measurement, inherently available in this measurement approach. For high precision FMCW measurements, usually either an electro-optical phase-locked-loop (EO-PLL) or a resampling method is necessary to reduce laser nonlinearity.

In [1] the first monolithically integrated optical phase-locked loop is presented, which is used for mode-locking of two lasers, nevertheless not used for LIDAR applications. In [2], a fully integrated analog EO-PLL is presented, which allows an rms

depth precision of 8 μm at a range of ± 5 cm from the range baseline with a maximum range window of 1.4 m. A continuous-time EO-PLL, which allows a faster control and a precision of 558 μm at 2 m distance is shown in [3]. By utilizing a resampling approach, in [4] a 5.5 μm precision is achieved for a range of 1.5 m. A 5.74 THz frequency excursion, provided by an array of distributed-feedback (DFB) lasers spread over the whole frequency range, allowed to achieve an outstanding measurement precision of 0.18 μm at a range of 1.4 m in [5], however requiring a very complex setup. In [1]–[4] and [5] only real-time information is used to provide appropriate feedback to the laser.

Another approach for achieving a linear frequency tuning is modulating the laser with a pre-distorted signal to compensate the nonlinearities. In [6] and [7], a model is used to approximate the ideal waveform. Nevertheless, it is hard to completely model the laser tuning behavior. Therefore, in [8] an iterative learning control (ILC) is implemented, where the laser tuning waveform is iteratively improved over many chirps. This allows a very linear laser frequency tuning. However, due to the very laboratory-based setup the iterative convergence process for obtaining the optimal waveform takes several minutes. This would be too long for many applications, especially if laser tuning characteristics change with temperature. In [9], a more advanced iteration algorithm was developed, which enables settling with only six iterations, many less compared to [8]. Nevertheless, the control loop still consists of lab equipment and a computation-intensive algorithm, what limits speed and ease of integration. In [10] an FPGA-based laser sweep linearization architecture consisting of a real-time and a pre-distortion part is presented, which allows much faster settling. Nevertheless, as the phase measurement resolution is limited by the system clock, the achieved phase error is relatively high.

In order to prove the feasibility of the pre-distortion approach for real world applications, we developed an EO-PLL consisting of a real-time path and a repetitive control path, which settles in fractions of a second and can handle fast changes in the laser tuning characteristics. Additionally, due to the use of a time-to-digital converter (TDC), a very precise phase error measurement is possible, which allows excellent laser linearity. Furthermore, except for the optical parts, this EO-PLL is realized on a small printed circuit board (PCB) and no additional lab equipment is necessary, what allows direct integration for many applications.

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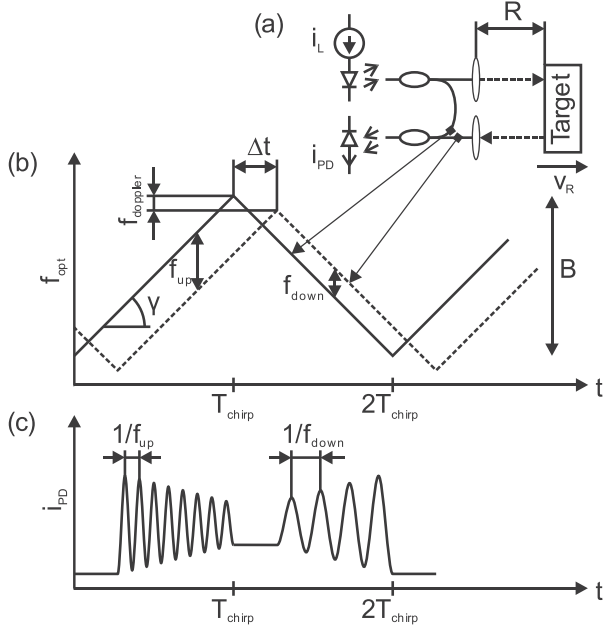


Fig. 1. Optical frequencies of measurement and reference arm over time.

A. FMCW LIDAR

In contrast to pulsed LIDAR, in FMCW LIDAR not the light intensity is modulated per se, but the optical frequency of the light f_{opt} is continuously chirped up and down as shown in Fig. 1(b). Usually, laser diodes are used as light source, as the light must be very coherent. Laser frequency modulation can be achieved in different ways, for example by tuning the laser injection current or by mechanically tuning the laser cavity length. Fig. 1(a) shows a simple FMCW LIDAR setup, where injection current tuning is used. A fiber optic coupler is used to split the laser light into two arms. One arm, the reference arm, is directly connected to a combining coupler. In the other arm, the measurement arm, the light is coupled out, is then reflected by a target and coupled back into the fiber again. Then both arms are combined, hence the light of the reference and the light of the measurement arm is superimposed and induces a phase-dependent photocurrent i_{PD} in a photodiode. Consequently, this setup forms a Mach-Zehnder interferometer (MZI). The solid curve in Fig. 1(b) shows the optical frequency coming from the reference arm, whereas the dashed curve shows the optical frequency coming from the measurement arm. The light intensity of the superimposed light of both arms oscillates with a beat frequency, which is the difference of both optical frequencies. This beat frequency is proportional to the arm length difference d of reference and measurement arm. The beat frequency is electronically measurable via the photocurrent i_{PD} as shown in Fig. 1(c). As the optical frequency chirp is achieved by tuning the laser current, this additionally generates a laser power modulation, which is also visible in Fig. 1(c). This is a side effect, which is not used in this work. The laser wavelength increases with rising injection current and consequently, the optical frequency decreases. Hence, in the up-chirp phase, laser

current, and power decrease, while they increase in the down-chirp phase. If the fiber lengths of reference and measurement arm are identical as in Fig. 1(a), the MZI arm length difference d equals two times the range R ($d = 2 \times R$), and can be derived with

$$d = 2R = \frac{c}{2\gamma} (f_{up} + f_{down}), \quad (1)$$

where c is the speed of light, γ the laser frequency modulation slope (see Fig. 1(b)) and f_{up} and f_{down} the corresponding beat frequencies for up- and down-chirp, respectively.

The ideal modulation slope γ can be calculated if the laser frequency excursion B , which is the total laser frequency change in one chirp, and the time duration T_{chirp} of a single chirp, are known:

$$\gamma = \frac{B}{T_{chirp}} \quad (2)$$

If the target is not moving, f_{up} and f_{down} are identical and d can be calculated with either f_{up} or f_{down} alone:

$$d = 2R = \frac{c}{\gamma} f_{up} = \frac{c}{\gamma} f_{down} \quad (3)$$

Yet, this is not the case if the target is moving with a velocity v_R as depicted in Fig. 1(a). Here, d changes with the arm length changing velocity v . As the optical frequency of the reflected light is Doppler shifted, the beat frequencies f_{up} and f_{down} differ by $2 \times f_{Doppler}$. This allows calculating v and v_R , which are proportional to $f_{Doppler}$, with [8]:

$$v = 2v_R = \lambda f_{Doppler} = \frac{\lambda}{2} (f_{up} - f_{down}) \quad (4)$$

II. FPGA-BASED EO-PLL WITH REPETITIVE CONTROL

As can be anticipated from Section I.A, the chirp linearity is of great importance for accurate distance measurement, as errors in f_{up} and f_{down} directly translate into a distance error. The modulation slope γ , which is shown in Fig. 1(b), has to be kept constant over the whole frequency excursion B . Although there exist some very linear tunable lasers, these lasers are usually relatively complex systems and not suitable or just too expensive for many applications.

In order to be able to use lasers with higher nonlinearities, a feedback loop is implemented. This is done by taking the beat frequency of a second reference-MZI with known arm length difference as feedback signal.

Fig. 2 shows the complete setup with both MZIs. The triangular waveform for the laser current tuning is generated with an integrator and the actual current slope, which is positive or negative depending on the actual chirp direction, is set by a Texas Instruments DAC8811 digital-to-analog converter (DAC). The DAC is controlled by an Intel Max 10 FPGA and a Schmitt trigger detects the crossings of the laser current thresholds. A balanced detector, consisting of two serially connected Hamamatsu S5972 PIN photodiodes with a bandwidth of 200 MHz and a following transimpedance amplifier (TIA) consisting of an Analog Devices ADA4625 operational amplifier with a chosen

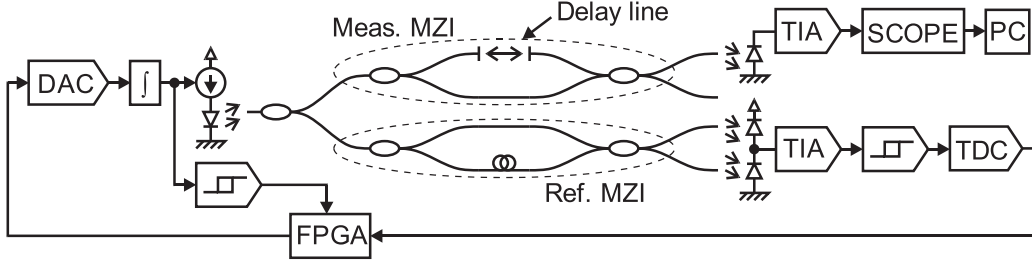


Fig. 2. Complete setup.

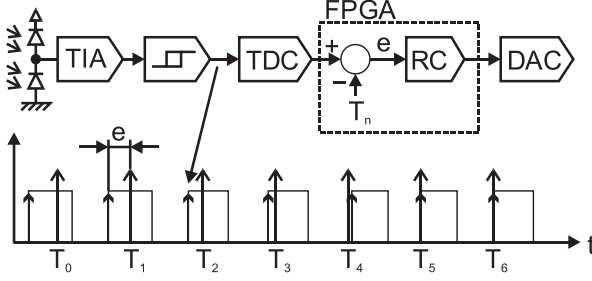
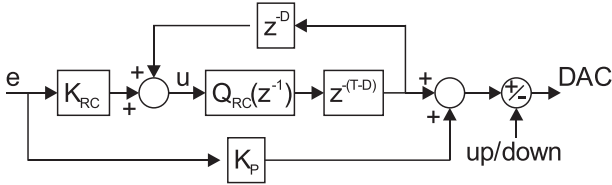
Fig. 3. Electronic part of feedback loop with phase error e getting smaller over time.

Fig. 4. Modified repetitive control structure.

transimpedance of 1 k Ω with adjustable bandwidth is used to receive the reference MZI feedback signal. A following Schmitt trigger transforms the beat signal into a square wave in order to get steep edges for the following ScioSense AS6500 TDC. Fig. 3 shows how the TDC allows to measure the absolute time of the beat signal's rising edges, which can be compared to the expected times T_n , which are calculated inside the FPGA based on the chosen beat frequency. The phase error e is then calculated for every period by subtracting the measured from the expected time value

$$e(n) = T_{TDC}(n) - T(n) \quad (5)$$

and is used as input for the repetitive control type controller, which sets the next DAC value. Fig. 4 shows the used repetitive control structure. It is basically a plug-in type repetitive control structure as described in [11], which is modified by putting the Q-filter in the forward instead of the feedback path of the repetitive part and by compensating the delay, mainly caused by TDC and DAC. There is a real-time feedback path via K_P and a repetitive control path via K_{RC} . In the repetitive control path, the values from the previous chirp are further corrected using the actual error e . For an up-chirp the values from the previous up-chirp and for a down-chirp the values from the previous down-chirp are used. T is the chirp period. Q_{RC} is a weighted-average filter

in the form

$$Q_{RC}(u(t)) = \alpha_1 u(t+1) + \alpha_0 u(t) + \alpha_1 u(t-1), \quad (6)$$

where $u(t)$ are the unfiltered input values and $\alpha_{1,2}$ are coefficients with $2\alpha_1 + \alpha_0 = 1$. The new calculated values are then used for the next chirp. As the whole EO-PLL system is not delay-free, mainly due to TDC readout and DAC write and set time, a time delay D has to be compensated. This is done by adding this delay in the feedback path of the previous value. The previously calculated value from the repetitive control path and the real-time feedback path are summed and depending on the chirp direction, the corresponding sign is added.

Due to the use of the integrator the DAC directly controls the modulation slope γ and the DAC resolution is also the maximum resolution of γ . Hence, a sufficient DAC resolution should be chosen which causes a negligible phase error e . In this work the used DAC has a resolution of 16 bits, which is by far sufficient for the performed measurements.

The modulation slope error $\delta\gamma$ is obtained with a cycle-to-cycle beat frequency measurement performed with the oscilloscope. Then with (3) we can determine the actual modulation slope $\gamma(t)$ with the actual measured beat frequency $f_{up}(t)$ or $f_{down}(t)$ and known MZI delay length difference d :

$$\gamma(t) = \frac{c}{d} f_{up}(t) = \frac{c}{d} f_{down}(t) \quad (7)$$

With this equation, we can calculate the actual modulation slope error $\delta\gamma$ by subtracting the constant ideal modulation slope γ_{ideal} :

$$\delta\gamma(t) = \gamma(t) - \gamma_{ideal} \quad (8)$$

Fig. 5 shows the modulation slope error $\delta\gamma$ over the chirp time as in [2] with a true triangular current modulation and with applied EO-PLL. One can clearly see the strong nonlinearity of the laser and the improvement with phase locking. Fig. 6 shows the corresponding DAC voltage, which is set by the control output.

In [8] and [9] a residual laser frequency error ν_{nl} is used as an indicator for chirp linearity. The actual laser frequency $\nu(t)$ consists of the carrier frequency ν_0 , an ideal linear frequency sweep part and a nonlinear sweep part $\nu_{nl}(t)$ [8]:

$$\nu(t) = \nu_0 + \gamma_{ideal}t + \nu_{nl}(t) \quad (9)$$

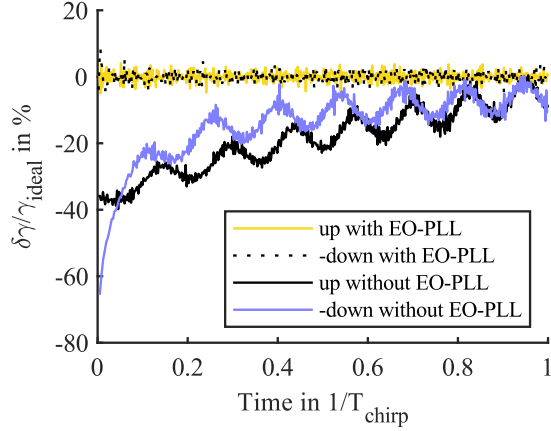


Fig. 5. Modulation slope error $\delta\gamma$ with and without EO-PLL over chirp time.

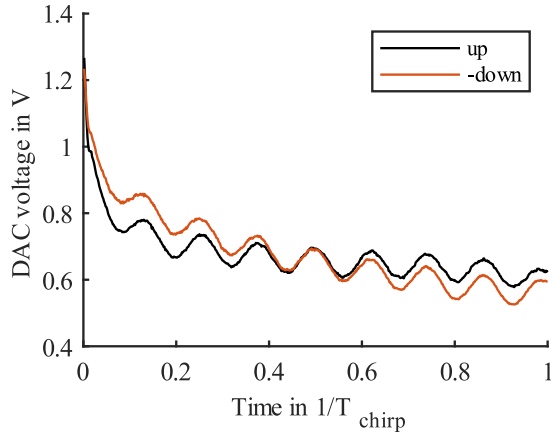


Fig. 6. DAC voltage over chirp time.

The actual laser frequency $\nu(t)$ can also be calculated by integrating the measured modulation slope $\gamma(t)$ over chirp time

$$\nu(t) = \nu_0 + \int_0^t \gamma(\tau) d\tau, \quad (10)$$

if the carrier frequency ν_0 is known. Combining (9) and (10) then allows to calculate the nonlinear sweep part $\nu_{nl}(t)$ with

$$\nu_{nl}(t) = \int_0^t \gamma(\tau) d\tau - \gamma_{ideal}t, \quad (11)$$

where $\gamma(t)$ is the measured modulation slope, which is expressed in (7).

This residual laser frequency error ν_{nl} with and without the proposed EO-PLL is shown in Fig. 7(a) and Fig. 7(b), respectively. The rms values of the residual error $\nu_{nl,rms}$ for up- and down-chirp are 0.72 GHz and 0.58 GHz without EO-PLL, and 1.59 MHz and 1.4 MHz with EO-PLL, respectively. The rms phase error, which is simply the rms value of the TDC-measured phase error e (Fig. 8) of a single chirp, is 2.84° for the up- and 2.68° for the down-chirp.

III. DISTANCE MEASUREMENT

The actual distance measurement was done with the upper measurement MZI in Fig. 2. In the reference arm, the 1×2

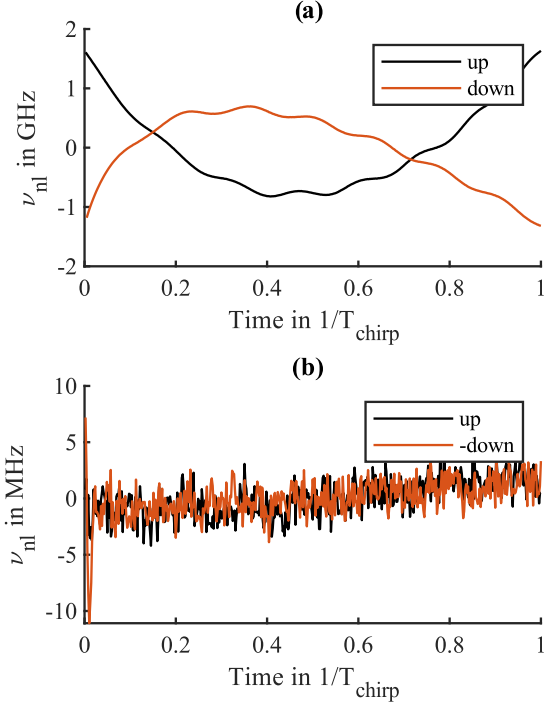


Fig. 7. Residual laser frequency error ν_{nl} over chirp time. (a) Without EO-PLL. (b) With EO-PLL.

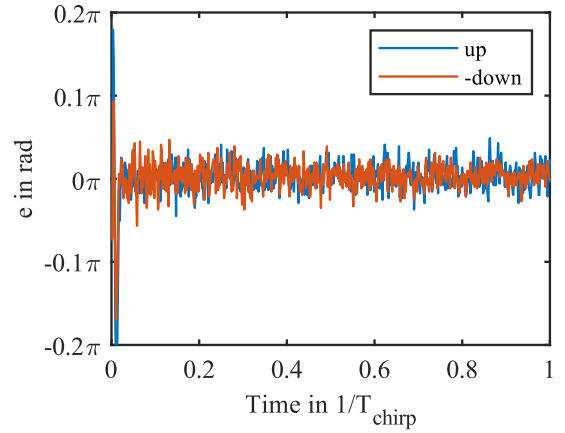


Fig. 8. TDC-measured phase error e over chirp time.

splitter and following 2×2 combiner are directly connected. In this work a Thorlabs LP780-SAD15 DFB laser with a mode-hop-free tuning range of 60 GHz, a linewidth $\Delta\nu$ of 2 MHz and a wavelength of 780 nm is used. In order to achieve a high frequency excursion B , the laser power is modulated between 3 mW and 12 mW. In the measurement arm, an OZ Optics ODL-300 adjustable fiber optic delay line with a resolution of $<1 \mu\text{m}$ and a travelling range of 100 mm is inserted to achieve real distance measurement. The beat signal is measured with a S5973 PIN photodiode from Hamamatsu with a bandwidth of 1 GHz and a responsivity of 0.51 A/W at 780 nm. A following HMC799 TIA from Analog Devices with a bandwidth of 700 MHz and 150 nA input rms noise amplifies the measurement signal for the following Teledyne LeCroy WaveRunner 204XI oscilloscope with a maximum sample rate of 10 GS/s, which

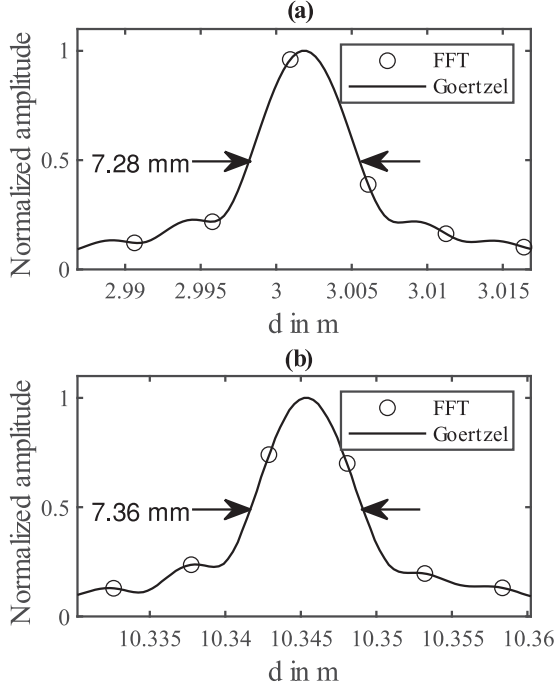


Fig. 9. FFT points and Goertzel spectrum calculated from oscilloscope data with sample rate of 10 MS/s around the peak frequency for (a) @ $d = 3.002$ m and (b) @ $d = 10.345$ m.

performs the analog-to-digital conversion. The PC controlling the measurement setup then reads out the oscilloscope. The following digital signal processing is done in LabVIEW.

As explained in Section I.A, the distance is proportional to the measured beat frequency. Hence, in order to find the corresponding peak in the frequency spectrum, the measured signal needs to be transformed into the frequency domain. One way to achieve this is by applying a simple fast Fourier transform (FFT). This method works for a fast and rough measurement but is not sufficient for high precision measurements as performed within this work. A much finer precision can be achieved by utilizing a Goertzel algorithm, which is generalized also for non-integer multiples of the fundamental frequency [12]. In order to keep computing time low, a first rough peak detection is done with an FFT. Then several spectral components (in this work 20) are calculated around the peak FFT bin with the generalized Goertzel algorithm. A much finer peak estimation is then possible by feeding the calculated Goertzel spectrum to the LabVIEW peak detector function, which also allows estimating the peak between calculated spectral components. Fig. 9(a) and Fig. 9(b) show FFT points and the calculated Goertzel spectrum around the peak frequency for a MZI delay length $d = 3.002$ m and $d = 10.345$ m. The full-width half-maximum (FWHM) of the peak stays around 7.3 mm over this range. Here 100 Goertzel points are calculated for a better visibility, but it is sufficient to use between 10 and 20 points for a good peak estimation.

IV. MEASUREMENT RESULTS

Different measurements were performed to show the functionality of the new EO-PLL scheme. First a sweep over the

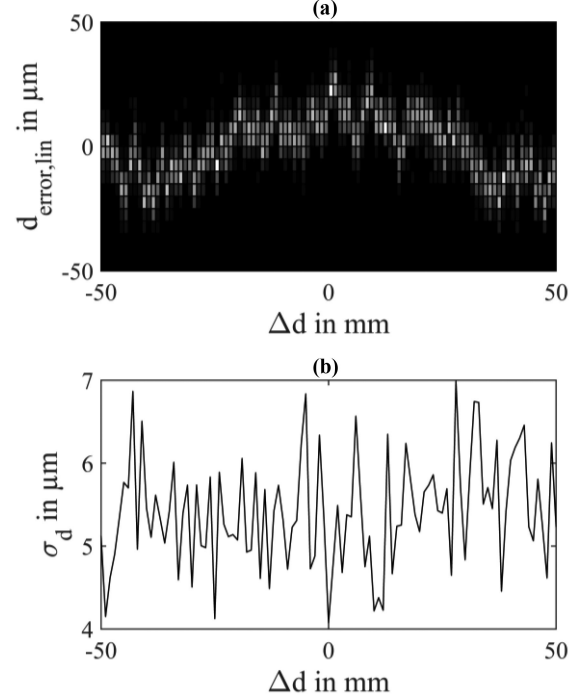


Fig. 10. Sweep over Δd from -50 to $+50$ mm @ $d = 3$ m with sample rate of 10 MS/s. (a) Histogram of linear errors $d_{error,lin}$. (b) Local standard deviation σ_d of 54 consecutive absolute measurements with a mean σ_d of 5.41 μ m.

whole delay line travelling range of 100 mm with a step size of 1 mm and a chosen oscilloscope sample rate of 10 MS/s was performed.

In order to remove the offset introduced by the fiber delay, a straight line is fit to the measured data as an offset adjusted ‘correct’ reference, and the error is calculated for every point by subtracting this reference from the corresponding measured distance values. This error does not contain the absolute offset error for the measurement distance d , nevertheless, this error can be calibrated easily. Fig. 10 shows a histogram of the distance errors $d_{error,lin}$ and the local standard deviation σ_d for 54 measurements per distance point at a normalized air MZI delay length of $d = 3$ m. The measured mean σ_d over the 100 mm is 5.41 μ m. We repeated this measurement also with a sample rate of 5 MS/s, 50 MS/s and 100 MS/s in order to identify the influence of the sample rate on the measurement precision, with a measured σ_d over the 100 mm of 5.61 μ m, 5.23 μ m and, 5.16 μ m, respectively. Hence, higher sample rates do not improve the precision considerably.

In order to increase d , an additional fiber was connected serially into the MZI measurement arm, which increased d to 10.3 m. Fig. 11 shows the corresponding results and the mean σ_d over the measured 100 mm is 16.35 μ m.

Doppler velocity measurement is also demonstrated in this work. Fig. 12 shows a continuous measurement of v and d , while the optical delay of the delay line increased by 8 mm. Here the absolute measurement d_{abs} is very noisy due to strong vibrations of the moving stage. In order to obtain the Doppler measurement accuracy, the velocity v (4) in Fig. 12(a) is integrated over time and is plotted as d_{int} . As being visible in Fig. 12(b), d_{int} gives

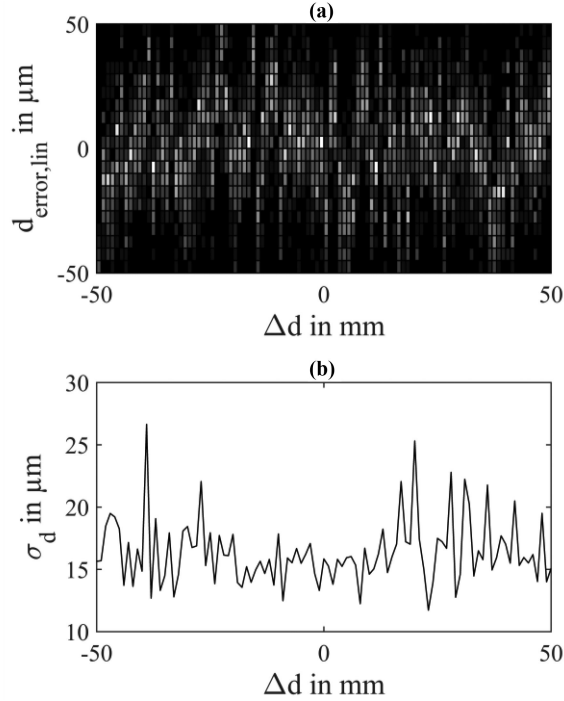


Fig. 11. Sweep over Δd from -50 to $+50$ mm @ $d = 10.3$ m with sample rate of 10 MS/s. (a) Histogram of linear errors $d_{error,lin}$. (b) Local standard deviation σ_d of 52 consecutive absolute measurements with a mean σ_d of $16.35 \mu m$.

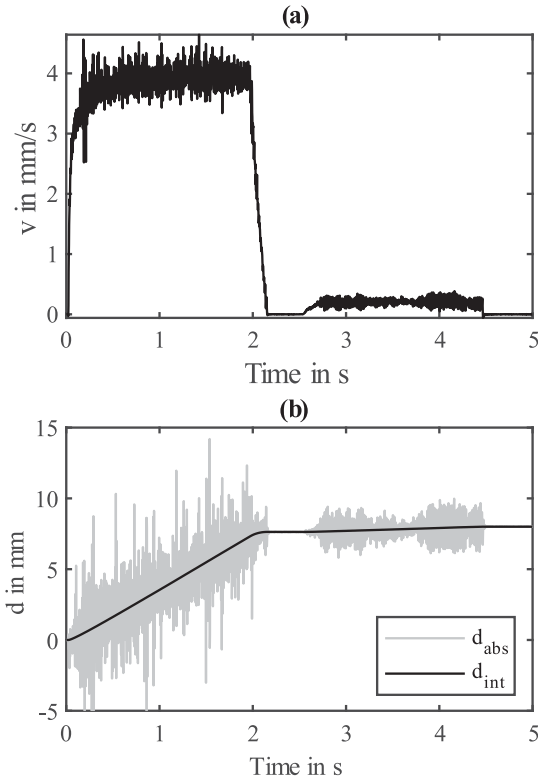


Fig. 12. Continuous measurement with 8 mm travelling distance and a sample rate of 2 MS/s. (a) Velocity v over time. (b) Absolute distance d_{abs} and integrated-velocity distance d_{int} over time.

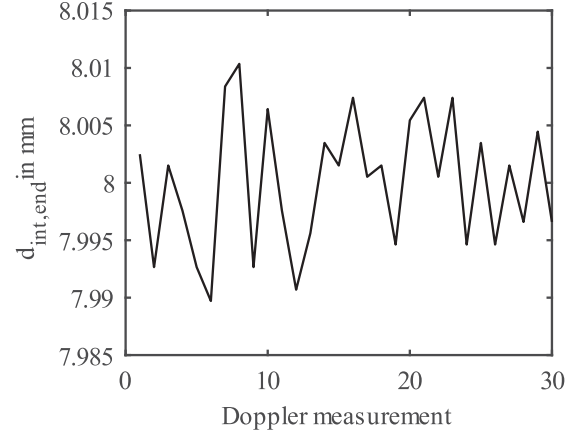


Fig. 13. Integrated-velocity distance d_{int} end values $d_{int,end}$ of 30 consecutive Doppler measurements with a repeatability of $5.78 \mu m$.

an almost perfect relative distance measurement. Fig. 13 shows the end values of the distance obtained by integrating v over time $d_{int,end}$ of 30 repeated measurements. The measured standard deviation of these 30 measurements is $5.78 \mu m$.

V. DISCUSSION

The main advantage of this EO-PLL is the possibility to use lasers with very nonlinear modulation slopes. The linewidth of the used laser $\Delta\nu$ is 2 MHz. The coherence length l_c can be calculated with [13]

$$l_c = \frac{2c \ln 2}{\pi \Delta\nu}, \quad (12)$$

where c is the speed of light. The resulting value for the used laser is $l_c = 66$ m.

A comparison to other state of the art results is shown in Table I. It should be mentioned that in the compared works the distance to a reflecting target is measured similar to Fig. 1(a). Performing the measurement like this, theoretically would further improve our distance measurement accuracy by a factor of 2. This is because in this work the total MZI delay length d is directly measured, whereas in the other works effectively a measurement of R , which equals $d/2$ (1), is performed, as there the light travels the measured distance two times.

Due to the TDC-based phase error measurement in combination with the FPGA, a continuous iterative and real-time adjustment of the laser driving waveform is possible, what represents a big improvement compared to [8] and [9] where this is not possible due to data transfer limits and high computing effort. This allows a waveform settling in fractions of a second in comparison to 10 min in [8] and probably at least several seconds in [9] (not explicitly mentioned). Additionally, the complexity of the electronic part of the control system is reduced to a small PCB (74 mm x 62 mm), which could be further reduced in size. Due to the use of a TDC in our work a much finer phase error measurement is possible, compared to [10], where the phase measurement resolution is limited by the system clock. In [10], the phase error e in the region of interest (ROI) can reach up to

TABLE I
COMPARISON

	This work	[2]	[4]	[5]	[8]	[9]	[10]
Method	EO-PLL/ Pre-distortion	EO-PLL	Resampling	Resampling	Pre-distortion	Pre-distortion	EO-PLL/ Pre-distortion
Laser	DFB	DBR	VCSEL	DFB array*	VCSEL/DFB	DFB	DFB
Wavelength	780 nm	1530 nm	1550 nm	1525 nm – 1565 nm	1550 nm	1550 nm	1550 nm
Precision σ_d / σ_R @ d / R^{**}	$\sigma_d =$ 5.88 μm @ $d = 3$ m	$\sigma_R = 8$ μm @ $R =$ ± 5 cm	$\sigma_R = 5.5$ μm @ $R = 1.5$ m	$\sigma_R =$ 0.18 μm @ $R = 1.4$ m	N/A	N/A	N/A
Measurement rate	0.53 k/s	180 k/s	10 k/s	1 k/s	4 k/s	1 k/s	1 kHz / 2 kHz
ADC sample rate	10 MS/s	N/A	500 MS/s	333 MHz	125 MS/s	N/A	125 MS/s
Chirp BW (B)	60 GHz	122 GHz	550 GHz	5.74 THz	155 GHz/ 36 GHz	26 GHz	35 GHz
Coherence length l_c	66 m	1.5 m	13 m	33.4 m	~ 10 m (VCSEL)	~ 132 m	660 m
Laser $(\delta\gamma/\gamma)_{\max}$	60 %	< 10 %	N/A	N/A	N/A	N/A	N/A
Laser freq. sweep nonlinearity $1-r^2$	$8.65 \cdot 10^{-9}$ up $6.7 \cdot 10^{-9}$ down	N/A	N/A	N/A	$1.8 \cdot 10^{-8}$ up $1.0 \cdot 10^{-8}$ down	$5.19 \cdot 10^{-8}$	N/A

*Relatively complex system; ** $d = 2 \times R$.

0.5π even after full settling, whereas in this work e reaches a maximum of 0.05π and an rms value of 0.015π (Fig. 8) in the ROI.

A value for the relative nonlinearity of the laser frequency sweep is the coefficient of determination r^2 of a linear regression analysis [8]. It should be as close to 1 as possible. Consequently, $1-r^2$ is ideally zero and is calculated with [8],

$$1 - r^2 = 12 \left(\frac{\nu_{nl,rms}}{B} \right)^2 \quad (13)$$

where $\nu_{nl,rms}$ is the rms value of the residual laser frequency error ν_{nl} which is shown in Fig. 7. The calculated values for $1-r^2$ are $8.65 \cdot 10^{-9}$ for the up- and $6.7 \cdot 10^{-9}$ for the down-chirp, which is a considerable improvement compared to the best values of [8] with $1.8 \cdot 10^{-8}$ for the up- and $1.0 \cdot 10^{-8}$ for the down-chirp and compared to the best value of [9], which is $5.19 \cdot 10^{-8}$. A further improvement is the extension of the region of interest (ROI), hence the linear part of the chirp which is usable for accurate distance measurement. In this work, the ROI is basically the whole chirp except for the first percent. This represents an improvement to [8] and [9], where 20% are outside of the ROI, hence the usable laser frequency excursion B is reduced by 20%.

Furthermore, we used a Goertzel algorithm generalized to non-integer multiples of the fundamental frequency in order to improve the measurement precision. As visible in Fig. 9 and Fig. 10, with an FFT resolution of 5.2 mm, a mean measurement precision σ_d of better than 6 μm for a continuous measurement

is possible, which is close to a factor of 1000 in comparison to a simple FFT peak measurement.

The main limitation of the presented EO-PLL is the relatively low chirp rate, which is determined by the TDC readout rate and the DAC sample rate. In this work, the TDC is read out by the FPGA via an SPI bus with a limited data rate. The chirp rate can be improved by using TDCs with an LVDS interface, what would increase the throughput by at least a factor of 10. Another possibility would be to implement the TDC into the FPGA. With these improvements and by using a faster DAC, a measurement rate of at least 10 kHz should be possible even with discrete components. These improvements will be part of our future work. An additional speed increase can be achieved by integration of the electronic part of the system on a chip. In addition, an integration of the photonic part is possible as shown in [2].

As a MZI is very sensitive to environmental influences, we would like to take these influences into account. For example, the carrier frequency of the laser will change with laser temperature, as the laser wavelength is also tunable by temperature with

$$\nu_0(T) = \nu_{0,T_0} + \gamma_T * (T - T_0), \quad (14)$$

where T is the temperature, ν_{0,T_0} the carrier frequency at temperature T_0 and γ_T is temperature modulation slope of the laser frequency, which has a maximum of about 30 GHz/K [14] for the used laser. The used current modulation slope γ is 64 THz/s. Hence, even if the laser temperature would change relatively fast, the influence on γ would be relatively small. Furthermore, as we

plan to further increase the chirp rate and hence γ , the influence on γ would further decrease. On the other hand, the error due to dispersion in the fiber or in the air might not be negligible for high precision measurements. Nevertheless, one can cope with this problem by temperature stabilizing the laser, which was done in our setup, also to keep the frequency excursion B constant for measurement comparability. Changing environmental temperature also causes a temperature change of the MZI. This is not negligible, as changing the MZI's temperature changes the fiber length and propagation speed in the MZI delay lines. One way to cope with this problem, is by building the fiber part of measurement and reference MZI as similar as possible. The fiber part of the delay lines of reference and measurement MZI should have identical lengths. The construction should guarantee that the temperature difference between both MZIs is small. Consequently, a changing environmental temperature effects both MZIs identically and a temperature drift in the absolute measurement is thereby compensated. If the application allows a temperature stabilization of laser and the total fiber setup, this can remove any temperature dependence of the measurement setup except for the free-space part.

An additional challenge are mechanical vibrations, which can strongly influence absolute distance measurement accuracies as already addressed for the Doppler measurements (Fig. 12) in Section IV. Nevertheless, these disturbances are caused by low frequency vibrations and can be strongly reduced by increasing chirp rates.

VI. CONCLUSION

We present a new digital EO-PLL with repetitive control for FMCW LIDAR applications, able to tune the laser frequency with almost perfect linearity, even with lasers with a very non-linear modulation slope. This approach allows choosing lasers for FMCW LIDAR applications with high coherence lengths without putting focus on modulation slope linearity. It furthermore represents a big improvement to previous works using a pre-distortion approach ([8], [9]) due to easier integration, much smaller waveform settling time and increased ROI, or much smaller phase error compared with [10]. The EO-PLL is tested with distance measurements, where a measurement precision of 5.41 μm is achieved for a MZI delay length of 3 m. Even for a MZI delay length of 10.3 m, a competitive precision of 16.35 μm is measured.

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